Ahmad Ghalawinji

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Profile

MSc graduate in Data Science and AI (ENSIMAG / Université Grenoble Alpes) with research experience in machine learning, large-scale data processing (Spark), feature engineering on multi-source observational datasets, and scientific computing. Experienced in end-to-end data analysis pipelines, controlled experiments, and scientific writing. Seeking a research position to apply computational and ML methods to scientific data analysis.

Education

MSc — Data Science and Artificial Intelligence (MoSIG M2) Sep 2023 – Sep 2024
ENSIMAG and Grenoble Alpes University *Grenoble, France*
Relevant coursework: *Advanced ML, Deep Learning, NLP, Scientific Computing*

MSc — Informatics (MoSIG M1) Sep 2022 – Sep 2023
ENSIMAG and Grenoble Alpes University *Grenoble, France*
Relevant coursework: *Algorithms, Numerical Methods, Research Methods*

BSc — Computer Engineering Sep 2016 – Sep 2021
Tishreen University *Latakia, Syria*
Relevant coursework: *Linear Algebra, Probability, Numerical Analysis, Data Structures*

Research & Professional Experience

AI Engineering Research Intern Feb 2024 – Aug 2024
Amadeus IT Group *Nice, France*

- Designed and implemented a full data analysis pipeline integrating three heterogeneous data sources (historical booking flows, travel timing records, competitor pricing) into a unified dataset of 150+ columns for ML-based demand forecasting.
- Processed large-scale observational datasets (>700 daily snapshots, 3–200 GB each) using Apache Spark on a distributed cluster, performing schema reduction (150 to 30 columns), temporal joins, and missing-value imputation across 26 origin-destination markets.
- Engineered 49 numerical features from competitor pricing data across multiple time windows and performed iterative feature selection guided by SHAP-value analysis, identifying a 7-feature subset that achieved the best forecasting improvement (34.3% RMSE reduction over the analytical baseline).
- Evaluated model performance using RMSE; interpreted results with SHAP to uncover learned temporal patterns (seasonal, day-of-week, booking-horizon effects). Authored a 50-page MSc thesis and presented findings to a scientific jury.
- **Technologies:** Python, Apache Spark, Databricks, Azure, MongoDB, SHAP, Scikit-learn, Jupyter, LaTeX.

Machine Learning Research Intern Feb 2023 – Jul 2023
Inria (French National Institute for Research in Digital Science) *Grenoble, France*

- Conducted research on deep generative models (variational autoencoders, transformers) for self-supervised representation learning, training and evaluating models in PyTorch on GPU clusters.
- Designed controlled experiments on latent-space representations with ablation studies, hyperparameter sweeps, and reproducible experiment tracking; wrote up results following academic standards.
- **Technologies:** Python, PyTorch, NumPy, Matplotlib, VAEs, GPU computing, LaTeX.

Software Engineering Research Intern Feb 2022 – Jul 2022
Laboratoire d'Informatique de Grenoble (LIG) *Grenoble, France*

- Designed a computational meta-model for automated planning based on Petri nets, creating formal representations of process states and transitions.

- Implemented a Domain-Specific Language layer translating high-level specifications into executable plans, enabling systematic exploration of solution spaces.
- **Technologies:** Python, Petri Nets, Formal Modelling, Automated Planning.

Selected Projects

Patient Similarity Explorer — Scientific Data Analysis & Visualization | |

- Built an analysis tool to explore similarity in high-dimensional clinical data by computing pairwise cosine similarity, constructing k-NN graphs, and applying t-SNE for dimensionality reduction and visualization.
- Implemented unsupervised methods to identify prototypical and atypical samples via representativeness and atypicality metrics, enabling case-based exploratory analysis over structured observational data.
- Delivered an interactive dashboard with cohort-level distribution analysis, nearest-neighbour retrieval, and graph visualization for interpreting complex multi-feature datasets.
- **Technologies:** Python, Scikit-learn, t-SNE, Cosine Similarity, Graph Analysis, Streamlit, Unit Testing.

Climate Change & Youth Fitness — Statistical Analysis & Forecasting | |

- Conducted quantitative analysis of climate and fitness datasets, engineering temporal and geographic features and applying XGBoost regression to model relationships between environmental variables and physiological outcomes.
- Produced scientific visualizations and a Streamlit dashboard to communicate findings, including correlation analyses, feature-importance plots, and forecast outputs.
- **Technologies:** Python, XGBoost, Pandas, Streamlit, Matplotlib, Seaborn.

Port Calls Detection — Large-Scale Geospatial Data Pipeline | |

- Built an automated ETL pipeline processing large volumes of vessel AIS positioning data to detect and classify port-call events, using geospatial algorithms (Haversine distance, spatial clustering) on time-series coordinate streams.
- Orchestrated batch processing with Apache Airflow and PostgreSQL/PostGIS; implemented filtering heuristics (speed thresholds, positional drift) to discriminate true events from false positives in noisy sensor data.
- **Technologies:** Python, PostgreSQL, PostGIS, Apache Airflow, Pandas, GeoPy, Docker.

Skills

Programming & Scientific Computing: Python, SQL, C++; NumPy, SciPy, Pandas, Matplotlib, Jupyter, PyTorch, Scikit-learn, XGBoost, SHAP
Data Engineering & Distributed Computing: Apache Spark, Databricks, Apache Airflow, Azure, PostgreSQL, PostGIS, MongoDB, Docker
Research & Communication: Experimental design, scientific writing (LaTeX), reproducible analysis, Git, Linux

Languages

Arabic: Native | **English:** Advanced (C2) | **French:** Intermediate (B2)

References

Gerardo Ayala Solano — Principal Specialist, Amadeus — gerardo.ayalasolano@amadeus.com
Simon Nanty — Principal Data Scientist, Amadeus — simon.nanty@amadeus.com
Nicolas Hili — Assistant Professor, Grenoble Alpes University — nicolas.hili@univ-grenoble-alpes.fr